

Complete Food and Home Cooking

Environmental and health evidence for two ways of eating a vegan diet

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Written with AI research and drafting assistance; see the note on sources and method at the end

Summary

A category of food exists now that did not exist fifteen years ago: powders formulated to be a complete diet on their own. Huel, Jimmy Joy, Mana, Soylent. Mixed with water they supply calories, protein, fat, fibre and the full set of vitamins and minerals in the amounts health authorities recommend, and most of the European brands are vegan. Part of the appeal is environmental — nothing spoils, portions are exact, no cooking is involved.

Whether that beats simply cooking plant food at home has never been tested in the published literature. It comes out close enough that environmental impact is not a good reason to pick either one. Both land somewhere between 1 and 3 kg of CO₂e per 2000 kcal, roughly one day of food for one adult. For scale, the same measure for a meat-heavy diet is about 10 kg [1]. Whether someone eats plants at all is a factor-of-four decision; which *form* the plants take is a factor-of-three decision at most, and much of that three is measurement rather than substance.

Four findings sit behind that.

There is almost nothing to check. No independent assessment of any complete food has ever been published in a peer-reviewed journal. One manufacturer publishes an absolute number; the rest publish either nothing or comparisons with no figures underneath them.

The footprint is created by refining the ingredients, not by making the product.

Whole soybeans carry 1.22 kg CO₂e per kilogram. Turned into protein isolate they carry 7.23 — about six times as much, with 84 per cent of it coming from the factory rather than the field [4]. Mixing the finished powders together, the step people picture when they imagine industrial food production, accounts for 0.2 per cent of the total. The assembly really is nearly free; the ingredients arrive expensive.

The vitamin blend is not the rounding error it is assumed to be. A generic premix carries 15 kg CO₂e per kilogram — more, per kilogram, than beef. At 2 per cent of the product by weight it still contributes roughly a sixth of the ingredient footprint, more than the oil and the rice flour combined [8].

On health, the evidence does not exist yet. Nobody has followed anyone living mainly on one of these products for longer than four weeks. The usual objection — that they are ultra-processed — does not transfer cleanly, because the health risks attached to that category concentrate in processed meat and sweetened drinks rather than in fortified plant products. Meanwhile the diet these products compete with has a documented weakness of its own: vegans who do not supplement carefully show markedly higher fracture rates.

What follows from all of this is that the format of the food is the wrong lever. Four things move the number more than the choice between a shake and a stew: how the cooking is

powered, how much food gets thrown away, how the shopping gets home, and whether the vegetables were grown in a heated greenhouse in winter.

1 The two things being compared

A **complete food** is a powder formulated to supply everything a person needs. It keeps for a year at room temperature, requires no cooking, and is normally bought online in bulk. The category began with Soylent in 2013 and now includes Huel in the UK, Jimmy Joy in the Netherlands, Mana in the Czech Republic, Saturo, YFood and others. The products examined here are all vegan.

The comparison is against a **regional, seasonal, whole-food vegan diet**: staples and produce bought in Germany, weighted toward what is in season or in domestic cold storage, and cooked at home.

Holding both sides vegan is what makes the question interesting. Removing animal products is well established as the dominant lever on dietary footprint, so fixing it leaves a narrower question: given that someone eats plants, does the form those plants arrive in change anything?

Manufacturers have an obvious commercial interest in the answer, so independent databases are preferred wherever both exist, and every manufacturer figure is labelled as one.

2 Why the numbers are hard to compare

Food footprints come from **life-cycle assessment** — an accounting method that traces every input, from fertiliser and diesel through factory electricity to packaging and transport, and adds up the emissions attributable to one unit of product. Two choices decide what such an assessment says, and both are easy to get wrong.

The first is the **system boundary**: where the accounting starts and stops. Cradle-to-gate stops when the product leaves the factory. Cradle-to-shelf continues to the supermarket. Cradle-to-grave includes cooking, refrigeration and waste at home. A product measured cradle-to-gate always looks better than the same product measured cradle-to-grave, so comparing across boundaries produces nonsense.

The second is the **functional unit**: what the footprint is expressed per. This one decides the answer here. Complete food is a dry powder at about 417kcal per 100g. Fresh vegetables are 85 to 95 per cent water. Compare them per kilogram and the powder looks roughly ten times better — entirely because of the water. A kilogram of powder is four days of food; a kilogram of tomatoes is a side salad. Everything in this paper is therefore expressed **per 2000 kcal**, and any manufacturer figure quoted per kilogram should be treated as uninformative until converted.

Three sources carry the quantitative work. **Agribalyse 3.1**, from the French environment agency, covers whole foods from cradle to shelf and breaks each figure down by stage — agriculture, processing, packaging, transport, retail. It is the only public database listing seasonal and out-of-season produce separately, which is what makes the seasonality question answerable at all [2]. **CarbonCloud ClimateHub** covers the industrial ingredients Agribalyse does not — protein isolates, oil powders, vitamin premixes — at factory gate, each split into agriculture, processing, transport and packaging [3]. And **Scarborough and colleagues**, in *Nature Food*, provide the measured reference point: the actual diets of 55,504

people, each food linked to a review of 570 life-cycle assessments covering 38,000 farms in 119 countries [1]. That is what people really eat, not a modelled ideal.

Household waste, cooking energy and the trip to the shop are added as separate, visible lines rather than folded into the food figure, so any comparison here can be re-run at a different boundary.

3 What the manufacturers publish

Very little, and almost none of it independently checkable.

Brand	Published product footprint	Verification
Huel	0.6 kg CO ₂ e per 400 kcal serving, 83 % of it from ingredients	Company data, assured to ISO 14064-3:2019; no methodology published
Jimmy Joy	None. Reports 103 t CO ₂ e offset in total	Offsetting, not measurement
Mana	None found	—
Soylent	Comparative ratios only, no absolute figure	Company-commissioned, unpublished
Saturo, YFood, Feed., Queal	None found	—

Huel is the outlier and deserves credit for it. The company measures its emissions annually at both company and product level and has the result independently assured [11, 10]. Its stated design target is below 0.5 kg CO₂e per meal, and its help centre describes the measurement as covering ingredients, manufacturing, packaging and delivery [12].

That assurance is nonetheless weaker than it sounds. The standard used verifies that an emissions inventory was compiled correctly. It does not require the company to disclose which background database it used, how emissions were divided between products, or where the system boundary was drawn. Without those three things an outsider cannot reproduce the figure or compare it fairly against a database value.

Jimmy Joy illustrates the more common pattern. Its sustainability page describes portion control, a twelve-month shelf life, partly recycled packaging and a tree-planting partnership, and states that the company has offset 103 tonnes of CO₂e in total — which the page itself equates to taking 22 cars off the road for a year [13]. None of that is a footprint. Offsetting is a payment made after the emissions have happened, and a cumulative 103 tonnes across the life of a company is far too small to cover a supply chain that grows and processes ingredients. There is no measurement there to examine.

The absence is the single most important methodological fact in this area: **no independent, peer-reviewed assessment of any complete-food product exists.** Every quantitative statement about these products, including the ones below, is either a manufacturer claim or a reconstruction from ingredient data.

4 Where the footprint actually comes from

What determines a formulated food's footprint is how heavily its ingredients were refined before they reached the mixer. The effect is large and it is measurable.

Refining means separating one component of a crop from the rest. Milling a grain into flour barely counts — the whole seed is still there, just smaller. Pulling the protein out of a bean

and discarding the rest is heavy refining: it needs water, chemicals to shift the acidity so the protein separates, and then a great deal of heat to dry the result back into a powder. That energy shows up directly in the footprint.

whole soybean	→	soy flour	→	soy protein isolate
1.22		0.64		7.23
kg CO ₂ e per kg of ingredient. 84 % of the last figure is factory energy, not farming.				

The same pattern repeats across every pair that has been measured.

Ingredient	kg CO ₂ e per kg	Share from processing
Soybean	1.22	≈0 %
Soybean meal	0.64	21 %
Soy protein isolate (90 % protein)	7.23	84 %
Rolled oats	1.20	2 %
Oat flour, whole grain	1.41	—
Hydrolysed oat flour	4.17	77 %
Sunflower oil	1.73	12 %
Sunflower oil dried into powder	3.13	45 %
Pea protein isolate (83 % protein)	3.62	60 %
Vitamin and mineral premix	15.0	100 %

All values are CarbonCloud verified product reports, at factory gate [4, 5, 6, 7, 8].

Grinding is close to free. Extracting a protein or drying an oil onto a carrier is not, and the extra cost is process energy rather than agriculture — which means sourcing the crop better does not remove it.

This has a consequence neither the manufacturers nor their critics tend to mention: **two products in the same category can differ from each other more than either differs from home cooking.** Jimmy Joy's Plenny Shake is built mainly from flours, with soy protein isolate appearing only fifth on the ingredient list [9]. Huel Powder is built around pea and rice protein isolates and a dried oil. By the table above, a flour-based formulation should sit near the bottom of the plausible range and an isolate-based one near the top. No brand publishes enough to confirm it.

The vitamin premix deserves separate attention, because the usual assumption about it is wrong. It is dismissed on the grounds that the quantities are tiny — micrograms of B12, milligrams of most of the rest. That is true of the vitamins and false of the premix, which is mostly mineral salts and carrier material and which at 15 kg CO₂e per kilogram is more carbon-intensive per kilogram than beef is per kilogram of meat [8]. At roughly 2 per cent of product mass it still contributes about 0.30 kg CO₂e per kilogram of finished powder. That is a line item, not noise.

5 Estimating a complete food from its ingredients

Because no product assessment exists, one has to be built. The method is straightforward, and it works because food labelling is regulated: the ingredient list must be in descending order by weight and the nutrition panel must be accurate. Between the two, find the set of ingredient quantities that reproduces the panel while respecting the ordering, then apply the emission factors above.

Plenny Shake declares, per 100 g: 417 kcal, 20.8 g protein, 15.6 g fat, 43.8 g carbohydrate and 8.6 g fibre, with the ingredient order oat flour, soy flour, sunflower oil, rice flour, soy protein isolate, ground golden flaxseed, inulin, vitamin and mineral mix [9]. The fitted composition reproduces protein, fat and fibre to within 2 per cent and carbohydrate to within 9.

Component	g/100 g	kg CO ₂ e/kg	Contribution	Share
Oat flour	52.9	1.41	0.747	40.0 %
Soy flour	30.4	1.40	0.426	22.8 %
Vitamin and mineral premix	2.0	15.0	0.300	16.1 %
Soy protein isolate	2.8	7.23	0.202	10.8 %
Sunflower oil	5.2	1.73	0.090	4.8 %
Rice flour	4.3	1.90	0.081	4.3 %
Ground flaxseed	0.8	2.40	0.020	1.1 %
Ingredients			1.87	100 %
Dry blending			0.005	
Multilayer pouch			0.12	
Parcel delivery, bulk order			0.15	
Total		2.14 kg CO₂e/kg		
Per 2000 kcal		1.03 kg CO₂e		

The central estimate is 1.03 kg CO₂e per 2000 kcal, with a range of 0.86 to 1.31 across the plausible span of each input.

The line worth pausing on is dry blending at 0.005 — two tenths of one per cent of the total. That step is the factory: powders going into a mixer and coming out combined. It is the part people picture when they imagine industrial food production, and it is almost free. Everything that mattered happened before the ingredients arrived.

5.1 A gap that needs explaining

This estimate is about a third of Huel’s self-reported figure, which works out to 3.0 kg CO₂e per 2000 kcal. The discrepancy is larger than the difference the whole exercise is trying to measure, so it cannot be left alone.

Part of it is the recipe. Re-running the same model on a Huel-like formulation — oats, pea protein isolate, flaxseed, rice protein, dried oil — gives roughly 1.4 to 1.5 kg CO₂e per 2000 kcal, because isolates and oil powders replace flours. That closes perhaps half the gap.

The most plausible explanation for the rest is that the figure is not measuring what it appears to measure. A company can count its emissions two ways: it can assess the product, or it can total up the whole business — offices, staff travel, marketing, servers, equipment — and divide by meals sold. The second number is always bigger. Huel reports carbon intensity per pound of revenue alongside the per-meal figure [11], and revenue intensity is a whole-business measure rather than a product one.

If that reading is right, the manufacturer’s number is conservative rather than flattering — which is unusual enough to be worth stating, because the reflexive assumption about company sustainability figures runs the other way. It remains a hypothesis; no published methodology would settle it. Both figures are carried into the comparison below.

6 The home-cooked side

Built the same way, from the same databases, so that the two sides can be compared without adjusting for method. Two days of food were constructed at 2000 kcal each: one optimised

around German seasonal and stored produce in winter, one reflecting an ordinary vegan shop including imported and greenhouse items.

	A: optimised regional and seasonal	B: ordinary imported + greenhouse
Food, cradle-to-shelf	1.14	1.47
Food thrown away at home	+0.14	+0.22
Cooking and refrigeration	+0.57	+0.57
Trip to the shop	+0.15	+0.30
As eaten, per 2000 kcal	1.99	2.55

Three results here matter more than the totals.

Seasonality is a large effect, and it is about heating, not distance. Agribalyse prices a seasonal tomato at 0.626 kg CO₂e per kilogram and an out-of-season one at 2.114 — a factor of 3.4 [2]. The stage breakdown shows where that comes from: the agricultural stage rises from 0.222 to 1.670 while transport barely moves, from 0.298 to 0.338. The penalty is the heated greenhouse. In basket B a single 200 g portion of out-of-season tomato contributes 0.43 kg CO₂e, which is the largest single line in the day and more than all the grain in it put together. Buying that tomato in summer instead of February takes about a fifth off the footprint of the whole day's food. Nothing else on the shopping list comes close.

Transport is a large share of a small number, which is why food miles mislead. For apples, transport is 64 per cent of the cradle-to-shelf footprint — but that footprint is 0.408 kg CO₂e per kilogram in total. For dried lentils it is 42 per cent of 0.582. The share is high precisely because growing plants is cheap; there is little else in the number for transport to compete with, so optimising it yields very little in absolute terms.

This reconciles two studies usually presented as contradicting each other. One found transport at 11 per cent of food-related emissions, with final delivery from producer to retailer at 4 [15]. A later one put global food transport near 19 per cent [16]. The higher figure counts all transport anywhere in the chain — fertiliser moving to farms, feed to livestock, machinery to dealers — rather than only finished food moving to a shop. For someone standing in front of two apples, the lower figure is the relevant one.

Cooking energy is the largest non-food line, and most published comparisons leave it out. At 0.57 kg CO₂e per day it is half the footprint of the food itself in basket A. It is also the one place where a powder mixed with cold tap water has an unambiguous structural advantage, along with waste: a sealed pouch with measured scoops does not go off at the back of a fridge.

For calibration against reality, the measured average UK vegan diet is 2.47 kg CO₂e per 2000 kcal, with a 95 per cent uncertainty interval of 2.09 to 3.36 [1]. That figure stops at the retail shelf, so it belongs beside the 1.14 and 1.47 rows rather than beside the totals, and it sits above both modelled baskets — which is expected, because real diets include plant milks, meat substitutes, coffee, chocolate, snacks and alcohol that a staples-only model leaves out. Germany's environment agency independently arrives at about 2.85 kg per day for a vegan diet with partly regional, seasonal and organic sourcing [14].

7 Side by side

Everything in the plant-based block lands between 1.0 and 3.0. The meat-heavy reference sits at 10.24. Moving from a meat-heavy diet to any vegan pattern is a factor of four; every choice available *within* the vegan block is a factor of three at most, and a good part of that

Option	kg CO ₂ e per 2000 kcal	Basis
Complete food, built up from ingredients	1.03 (0.86–1.31)	Reconstructed from verified ingredient factors
Optimised regional and seasonal winter day	1.99	Agribalyse plus German grid
Ordinary vegan day	2.55	Agribalyse plus German grid
Measured average UK vegan diet	2.47 (2.09–3.36)	Scarborough 2023, stops at retail
Complete food, manufacturer figure	3.00	Company figure, methodology unpublished
<i>For scale: measured UK high-meat diet</i>	<i>10.24 (7.04–15.95)</i>	<i>Scarborough 2023, stops at retail</i>

three is measurement rather than substance.

Two rows are not strictly comparable and are marked as such. The measured figures stop at the retail shelf while the two baskets and the reconstruction run through to consumption. Comparing like with like, the food-only numbers are 1.14 for the optimised basket, 1.47 for the ordinary one, 2.47 for the measured average and about 0.90 for the complete food. On that basis the complete food is ahead — but by a margin comfortably inside the uncertainty of a reconstruction that had to infer its own recipe.

8 Impacts other than carbon

Carbon is the only indicator any complete-food manufacturer reports. None publishes land use, water scarcity, nutrient pollution or biodiversity impact. That is a gap, not a clean bill of health.

What can be said from ingredient data is directional. The formulation is dominated by oats and soy, which are land-hungry per kilogram compared with vegetables but efficient per calorie. Agribalyse gives oat flakes 214 m²·year of land per kilogram against carrots at 9.54 — but oats deliver about 380 kcal per 100 g and carrots about 40, so per calorie the ranking reverses [2]. This is the functional-unit trap from Section 2 running in the other direction, and a reminder that a single number without its unit is not evidence of anything.

Two effects favour the complete food structurally and are not speculative. Household waste for a shelf-stable powder in pre-measured servings is effectively zero, against fresh fruit and vegetables making up around a third of avoidable household food waste in Germany. And packaging per calorie is low, because the product is dehydrated — although multilayer laminate pouches are in practice not mechanically recycled in Germany and go to energy recovery, a circularity failure the carbon number does not capture.

One effect runs the other way. Ready-to-drink versions ship roughly five times the mass per calorie, because they ship water, and they need rigid bottles rather than flexible pouches. The choice between powder and ready-to-drink within the category plausibly matters more than the choice between the category and cooking.

9 Health

The environmental question comes out close. The health question is genuinely unresolved, and the reason is an absence of evidence rather than a balance of it.

9.1 What is not known

No trial or cohort study has followed healthy, weight-stable adults getting most of their calories from a complete food for longer than four weeks. The longest published study is a single-arm, industry-funded pilot with twenty people who finished it.

The large body of research on meal replacements does not fill that gap. Those trials use meal replacements for deliberate weight loss in people with obesity or type 2 diabetes — a different intervention, in a different population, at a different calorie level.

Specific unknowns worth naming rather than glossing over: long-term effects on teeth and chewing muscles from a mainly liquid diet, appetite regulation over months, what low fibre variety does to the gut microbiome, and how well minerals are actually absorbed from an oat and soy matrix, which is high in phytate — a compound in seeds and grains that binds iron and zinc and reduces uptake. That last question has never been measured in these products.

9.2 The ultra-processed food question

Complete foods fall squarely into the category most nutrition research now calls ultra-processed. That category has an unfavourable reputation, and at first glance the evidence supports it: a 2024 review pooling 45 analyses found higher intake associated with a 21 per cent higher risk of dying from any cause, a 50 per cent higher risk of cardiovascular death, and raised risks of type 2 diabetes and depression [17]. Only two of those estimates reached moderate certainty on the standard grading scale; most remained low or very low, which is normal for observational nutrition research.

The problem is what the category contains. Processed meat, soft drinks, confectionery and a fortified vegan shake all sit in the same bucket, and a large European study shows they do not behave alike. Across 266,666 participants followed for a median of 11.2 years, higher intake did raise the risk of cancer and cardiometabolic disease — but the effect concentrated in animal-based products and sweetened drinks, while ultra-processed breads and cereals and plant-based alternatives showed no detectable association at all [18]. Complete foods belong to the second group by composition.

The strongest evidence on the other side is the only randomised trial that isolates processing itself. Twenty weight-stable adults lived in a research ward and ate an ultra-processed diet for two weeks and an unprocessed one for two weeks, in random order. The two menus were matched for calories offered, energy density, protein, fat, carbohydrate, sugar, salt and fibre, so the only difference was processing. On the ultra-processed diet people ate 508 kcal a day more without being asked to and gained 0.9 kg; on the unprocessed diet they lost 0.9 kg [19].

That is causal evidence that processing drives overeating regardless of nutrient content. How far it transfers is arguable in both directions — the menu in that trial was conventional packaged food, not a fortified formulation, but the mechanism identified, eating faster with less satiety per calorie, applies at least as strongly to something drinkable. Nobody has run the experiment with a complete food.

9.3 The risk on the other side

It is easy to treat home-cooked vegan food as the safe default. The data does not entirely support that.

In a cohort of at least 54,898 people followed for an average of 17.6 years, vegans had 2.31 times the hip-fracture rate of meat eaters, with a 95 per cent confidence interval of 1.66 to

3.22. In absolute terms that is 14.9 additional hip fractures per 1000 people over ten years. Total fracture risk was also raised, at 1.43 [20]. The associations weakened but persisted after adjusting for dietary calcium and protein, which points at intake adequacy rather than at anything intrinsic to plant food.

Micronutrient status tells the same story from another angle. Vitamin B12 is the familiar case; iodine, selenium, zinc and long-chain omega-3 fats are the less-discussed ones, and in European populations — where soils are low in selenium and salt is not routinely iodised — they are not rare deficiencies.

This is the honest case in favour of complete food, and it is stronger than the environmental one. A product supplying calcium, vitamin D, B12, iodine, selenium, zinc and algal omega-3 at reference levels removes, by construction, the exact failure the fracture data points at. Whether it does so in practice has never been tested, and “should eliminate” is not “eliminates”.

9.4 Contaminants

Plant proteins concentrate heavy metals from soil more than animal proteins do, and organic sourcing tends to increase rather than reduce cadmium and lead uptake. Exceedances of California’s Proposition 65 thresholds are common in commercial protein powders, though those thresholds are far stricter than European limits and presence at those levels is not evidence of harm.

The consideration is real for a product eaten daily in quantity rather than occasionally, and it is a place where the varied diet has a genuine advantage: eating many different plants dilutes any single contaminated input.

10 The strongest objection

The case against the conclusion runs like this. The comparison flatters complete food, because the home-cooked baskets carry cooking, shopping and waste while the powder carries none of them. Those three lines are 0.86 of the 1.99 kg optimised total — nearly half. Strip them out and the food itself is 1.14 against about 0.90, close to parity and well inside the error bars of a reconstruction built on an inferred recipe.

That objection is correct, and it is the conclusion rather than a refutation of it. The lines are not an artefact; they are real emissions a powder user genuinely does not produce. But their size relative to the food is the point. What separates these two options is not the food, it is everything around the food. Someone who cooks on an induction hob with renewable electricity, shops by bicycle and wastes nothing has already closed the gap without changing what they eat.

A second objection is that the reconstruction is the weakest number here and is doing the most work. Also correct. The recipe is inferred rather than disclosed, the emission factors come from a commercial database rather than peer review, and the only manufacturer figure available disagrees with it threefold. It is presented as the better estimate because its inputs are individually checkable and the manufacturer’s are not. A published product assessment would supersede it immediately.

11 Established, likely, unknown

Established. Refining, not manufacturing, drives the footprint of formulated foods: soy protein isolate costs about six times its parent bean, 84 per cent of that from processing. Seasonality is a factor of 3.4 for greenhouse-sensitive produce, driven by heating rather than distance. Transport is a large share of a small absolute number for plant foods. Cooking energy is a major line and is usually omitted. Vegan diets of any construction come to roughly a quarter of a meat-heavy diet's footprint. Vegans who do not supplement carefully carry a measured, substantial excess fracture risk.

Likely. Complete food probably has a modestly lower as-eaten carbon footprint than home cooking, driven mainly by zero cooking energy and zero waste rather than by the food itself. The vitamin premix is a real contributor at roughly a sixth of the ingredient footprint. Flour-based formulations are materially lighter than isolate-based ones. The health risks attached to ultra-processed food probably do not transfer wholesale to fortified plant formulations — though this rests on subgroup analysis rather than direct study.

Unknown. Any peer-reviewed assessment of a complete-food product. The system boundary behind the one published manufacturer figure. Land, water, nutrient-pollution and biodiversity impacts of any of them. Health outcomes beyond four weeks. How well minerals are absorbed from a high-phytate formulated matrix. Whether fortification removes the fracture risk it appears designed to remove. Long-term effects on the gut microbiome, appetite and oral health.

12 Conclusion

Environmental impact is not a good basis for choosing between these two ways of eating. Both sit in the 1 to 3 kg CO₂e band per 2000 kcal, and the levers that actually move the number — avoiding out-of-season greenhouse produce, not wasting food, not driving to the shop, and how the cooking is powered — cut across both.

The intuition that centralising production and adding all the vitamins at once must be efficient is half right. The assembly really is nearly free: mixing the powders is 0.2 per cent of the product's footprint. But the ingredients arriving at that mixer have been refined, and refining is where the energy went. Central manufacturing does not escape that cost; it depends on it.

On health the position is not a tie but a blank. There is no long-term evidence in either direction, and reading that blank as reassurance is as wrong as reading it as alarm. What is documented is that the whole-food vegan diet has a measurable failure mode — micronutrient adequacy, with fracture risk as its visible endpoint — that a fortified product is designed to solve and has never been shown to solve.

Conditionally, then. Where the motivation is environmental, the evidence does not support switching, and the same effort spent on seasonality, waste and cooking energy would achieve more. Where the motivation is convenience, cost or reliable micronutrient coverage, the environmental evidence neither supports nor blocks it — and partial substitution captures most of the waste and cooking-energy advantage while keeping the dietary variety whose long-term value is, unlike most of the above, not seriously disputed.

Appendix: basket composition

Both baskets normalised to 2000 kcal. Emission factors are Agribalyse 3.1 cradle-to-shelf values in kg CO₂e per kg; the last column is the resulting per-day contribution. Waste, cooking and shopping are added on top of these subtotals as shown in Section 6. Cooking assumes 1.6 kWh per day at the German grid factor of 0.353 kg CO₂e per kWh.

A: optimised regional and seasonal winter day				B: ordinary day, imported and greenhouse			
Item	g	Factor	kg CO ₂ e	Item	g	Factor	kg CO ₂ e
Potatoes	350	0.788	0.276	Tomato, off-season	200	2.114	0.433
Rye/wheat bread	180	0.760	0.137	Rice, dry	80	2.015	0.165
Oat flakes	100	1.220	0.122	Tofu	150	1.000	0.154
Kale	150	0.713	0.107	Banana	150	0.909	0.140
Walnuts	25	4.110	0.103	Bread	150	0.760	0.117
Sauerkraut	100	0.700	0.070	Walnuts	25	4.110	0.105
Rapeseed oil	25	2.500	0.062	Oat flakes	80	1.220	0.100
Apple, stored	150	0.408	0.061	Cucumber	150	0.512	0.079
Leek	100	0.611	0.061	Strawberry, off-season	100	0.592	0.061
Carrots	150	0.396	0.059	Sunflower oil	25	2.358	0.060
Beetroot	100	0.408	0.041	Chickpeas, dry	60	0.899	0.055
Lentils, dry	70	0.581	0.041				
Subtotal			1.14	Subtotal			1.47

Data and method. Agribalyse 3.1 queried through the ADEME data API, July 2026. CarbonCloud ClimateHub product reports and manufacturer pages read directly, July 2026. Recipe fitting by constrained random search over the simplex, minimising squared relative error against the declared nutrition panel subject to the declared ingredient ordering. All effect sizes and confidence intervals were checked against the primary publications rather than against secondary summaries. Uncertainty ranges reflect the span of published emission factors for each ingredient, not a formal propagation. Research and drafting were carried out with assistance from Claude (Anthropic); every source cited was retrieved and checked against the primary document, and responsibility for the content rests with the author.

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